

N-PORT Filing Pipeline & Runbook

How the tool works, what the audit found, and what to do next — step by step.

Period	2026-06 (quarter ending 30 June 2026)
Funds	105 in custodian · 104 building · 1 cash-only
Status	All output marked TEST — nothing filed
Branch	humanreview-live-gate
Prepared	30 July 2026

Before you start

Both source extracts stop on **24 June**, but the filing covers through **30 June**. Nothing downstream can be trusted until they are re-pulled. That is Step 1.

1 · How the pipeline works

Three files in, one XML per fund out. Four commands, plus one trip to a Bloomberg terminal.

YOU SUPPLY

- **Custodian file** — every position, all funds (US Bank) → `data/custodian/`
- **AP order book** — every create / redeem order → `data/orders/`
- **EagleSTAR export** — the accountant's daily snapshots → `data/fund_accounting/`



1

nport masters
COMMAND

Reads all three inputs and builds **two Excel workbooks** covering every fund at once — one for holdings, one for fund-level numbers. Also writes the **provenance** and **reconciliation** reports.



2

Bloomberg terminal
MANUAL

Open both workbooks, let the `=BDP()` formulas calculate, save as `.xlsx`. This fills company IDs, country, returns and bond durations.



3

**nport
mergehumanreview**
COMMAND ×2

Lists whatever has no feed in a review spreadsheet. You fill the blanks; running it again folds your answers back in, tagged as **human-reviewed**.



4

nport split
COMMAND

Pure copying. Chops the workbooks into per-fund files. No decisions made here — which is why you **edit the workbook, never the fund files**.



5

nport build
COMMAND

Sorts each row into a type, validates every required field, checks the LIVE gate, writes the XML and validates it against the SEC schema.



YOU GET

- `output/<TICKER>_2026-06.xml` — one per fund, ready for EDGAR

The rule the design follows

Every number has exactly **one owner**. A second source never writes — it only cross-checks into the reconciliation report. Where nothing has a feed, a human fills it in and it gets tagged.

WHAT	WHO OWNS IT
Positions, share counts	Custodian
Company IDs, country, returns, durations	Bloomberg
Dollars — net assets, gains, flows	EagleSTAR accounting
Anything with no feed	Human review spreadsheet

2 · What the audit found

The plumbing is sound. All 104 files pass the SEC schema. The problems are in the data going in.

The one root cause — stale source data

Both extracts stop on **24 June**; the period ends **30 June**. This produced **238 warnings across 137 of 152 funds**, leaving only 15 clear to file.

Most of those warnings are harmless. Accounting books a trade when it *settles*; the order book records it when it is *placed*. So a trade near a month boundary lands in different months in the two systems. Add up the whole quarter and they match exactly — **232 of 304** fund/side series tie precisely.

But 16 funds are genuinely wrong

They had accepted redemptions on **24 June** that had not settled by the time the accounting snapshot was taken. Accounting is the source for flows, so those filings report **zero redemptions**. Example: XIWC redeemed **\$374,892** and the file says \$0.00.

Two code defects — now fixed

DEFECT	BEFORE	AFTER
Percentages did not tie to the total. Each holding's % came from the custodian's fund size, but the filing reports the accountant's fund size. Two denominators.	85 / 104	104 / 104
Swaps never said long or short. The field was hardcoded blank even though the direction is decoded from the ticker two lines earlier.	0 / 58	58 / 58

Both fixed, tested and pushed. The percentage check is now enforced on every build — a mismatch fails the build instead of shipping quietly. Test suite: **656 passing**.

Smaller items

ITEM	ASSESSMENT
70 option deltas blank across 18 funds	No feed exists — comes from your risk system
GPTZ will not build	Correct — it holds only a \$125k cash seed, nothing to report
Monthly returns blank for most funds	Not a gap — verified against fund inception, 104/104 consistent
7 funds show no designated index	Fill in the review spreadsheet if required
OVERVIEW.md is out of date	Predates the review step and the accounting integration

3 · What to do — step by step

Work down the list. Do not skip Step 3 — everything after it inherits the answer.

1 Get a new EagleSTAR export through 30 June

Ask for it to be re-cut with a 30 June cutoff. Save the zip into:

```
data/fund_accounting/
```

2 Get a new AP order book through 30 June

Re-export it and save as:

```
data/orders/2026-06_orders.csv
```

The custodian file is already period-end. Leave it alone.

3 Check the dates actually moved

```
nport masters --dry-run
```

If either file still stops at 24 June, **stop here**. Everything downstream inherits the problem.

4 Build the workbooks

```
nport masters
```

Look for **flows from EagleSTAR TB + EagleSTAR gains/liabilities**. If it says *no fund accounting*, the zip was not found — stop and fix that first.

5 Read the reconciliation report

```
data/master/reconciliation_2026-06.csv
```

Most warnings should now be gone. Three things to look for:

- Warnings that cancel out across two months → ignore, that is settlement timing
- A fund with order-book redemptions but zero in accounting → investigate
- **FDRS** and **FDRX** → had real \$31M / \$9.5M gaps unrelated to dates

6 Go to the Bloomberg machine

Open both workbooks, wait for them to finish calculating, then save and close Excel.

```
data/master/security_master.xlsx  
data/master/filing_master.xlsx
```

Keep them as **.xlsx** — do not "save as CSV".

7 Create the review spreadsheet

```
nport mergehumanreview
```

8 Fill in the blanks

```
data/humanreview/2026-06_review.xlsx
```

Mainly the option deltas for the 18 defined-outcome funds. Save and close.

9 Merge your answers back in

```
nport mergehumanreview
```

Yes — the same command twice. First run creates it, second run absorbs your edits.

10 Push out to the funds

```
nport split
```

11 Test the build, then build for real

```
nport build --dry-run    # writes nothing, just reports problems
nport build
```

Expect **Built 104/108**. GPTZ failing is correct — it holds only cash.

12 Check the output

```
python scripts/verify_output.py
```

You want **CLEAN 104/104**. Then open two or three XMLs in `output/` and eyeball the fund name, period and net assets.

13 Go live

Set `liveTestFlag` to `LIVE` in `filing_master.xlsx`, save, then:

```
nport split
nport build
```

14 Upload the XMLs to EDGAR

If the build refuses to go LIVE

That is the safety gate doing its job — it blocks any fund whose reconciliation still has an unresolved warning. **Do not switch it off.** If it blocks nearly everything, the gate needs retuning to ignore the harmless month-shuffle warnings; ask before overriding.

Right now

Do Steps 1 and 2. Nothing else can move until those two files arrive.

4 · Where every data point comes from

Generated from the code by `scripts/field_provenance.py`, not written by hand. Every row cites the file and line that sets it, and the script refuses to run if a claim no longer matches the code.

Custodian Bloomberg EagleSTAR Human review Manual (workbook) fund_config.txt Tool constant
Derived

One writer per field

Each row below has exactly one source. A second system may *check* a value — the AP order book cross-checks EagleSTAR's flows, the custodian cross-checks net assets — but a cross-check never writes the filing.

Disagreements go to `reconciliation_2026-06.csv` and block a LIVE filing until resolved.

FIELD	SOURCE	HOW IT IS OBTAINED	CODE REFERENCE
FUND IDENTITY			
cik	fund_config.txt	Typed once when the fund launches	src/nport/config.py:209
seriesId	fund_config.txt	From the SEC series guide spreadsheet	src/nport/config.py:209
classId	fund_config.txt	From the SEC series guide spreadsheet	src/nport/config.py:209
seriesName	fund_config.txt	Typed once when the fund launches	src/nport/config.py:209
seriesLei	fund_config.txt	Typed once (GLEIF-issued)	src/nport/config.py:209
regName / regCik / regLei	fund_config.txt	Trust-level, same for every fund	src/nport/config.py:209
regStreet1 ... regZip / regPhone	fund_config.txt	Trust address, set once	src/nport/config.py:209
signerName / signerTitle / signerOrg	fund_config.txt	Who signs the filing	src/nport/config.py:209
HOLDING IDENTITY			
name	Custodian	SecurityName, truncated to 30 chars	src/nport/custodian.py:245
title	Custodian	SecurityName verbatim	src/nport/custodian.py:247
cusip	Custodian	Copied via =custodian! cell reference; foreign CINS → N/A	src/nport/custodian.py:671
ticker	Custodian	StockTicker column	src/nport/custodian.py:226
isin	Bloomberg	=BDP(bbgid, "ID_ISIN")	src/nport/master_sheet.py:152
lei	Bloomberg	=BDP(bbgid, "LEGAL_ENTITY_IDENTIFIER"); N/A if the issuer has none	src/nport/master_sheet.py:153

invCountry	Bloomberg	=BDP(bbgid, "CNTRY_OF_DOMICILE") — no default, a blank is a real gap	src/nport/master_sheet.py:154
POSITION			
balance	Custodian	Shares / par from the custodian	src/nport/custodian.py:676
units	Tool constant	Set by holding type: NS shares, PA par, NC contracts	src/nport/custodian.py:677
curCd	Tool constant	USD — the book currency for every fund	src/nport/custodian.py:641
valUSD	Custodian	MarketValue. Swaps override to the EagleSTAR mark-to-market	src/nport/custodian.py:642
valUSD (swaps)	EagleSTAR	PVal Total Unreal G/L Base — the MTM, never the notional	src/nport/custodian.py:1051
pctVal	Derived	valUSD ÷ the netAssets the filing reports (not the custodian's)	src/nport/custodian.py:1053
payoffProfile	Custodian	Sign of the share count; for swaps the ticker's L/S code	src/nport/custodian.py:725
assetCat	Tool constant	Set by classification: EC / DBT / DE / STIV	src/nport/custodian.py:493
issuerCat	Tool constant	Set by classification: CORP / UST / RF / OTHER	src/nport/custodian.py:679
fairValLevel	Tool constant	1 for listed, 2 for derivatives	src/nport/custodian.py:680
isRestrictedSec / isCashCollateral / isNonCashCollateral / isLoanByFund	Tool constant	N for every ETF holding	src/nport/custodian.py:645
DEBT (C.9)			
maturityDt	Bloomberg	=BDP(cusip Govt/Corp, "MATURITY")	src/nport/master_sheet.py:171
annualizedRt	Bloomberg	=BDP(cusip Govt/Corp, "CPN")	src/nport/master_sheet.py:172
couponKind	Bloomberg	=BDP(... "CPN_TYP"), mapped to the N-PORT enum	src/nport/master_sheet.py:173
isDefault / areIntrstPmntsInArRs / isPaidKind	Tool constant	N — no defaulted debt held	src/nport/custodian.py:760
OPTION			
putOrCall	Custodian	Parsed from the position name (trailing C / P)	src/nport/custodian.py:520
exercisePrice	Custodian	Parsed from the position name (strike)	src/nport/custodian.py:697
expDt	Custodian	Parsed from the position name (expiry)	src/nport/custodian.py:699
writtenOrPur	Custodian	Sign of the contract count	src/nport/custodian.py:700

shareNo	Tool constant	N/A — index options are on an index, not a share count	src/nport/custodian.py:695
delta	Manual (workbook) NO FEED	FLEX options do not price on Bloomberg. Typed into security_master.xlsx and preserved across rebuilds — from your risk system	src/nport/master_sheet.py:585
refIndexName / refIndexIdentifier	Human review	option_index sheet → applied at split	src/nport/master_sheet.py:786
counterpartyName / counterpartyLei	Tool constant	The Options Clearing Corporation — every listed/FLEX option clears through it	src/nport/custodian.py:91

SWAP			
swapFlag	Tool constant	Y for every TRS	src/nport/custodian.py:732
terminationDt	Custodian	Parsed from the swap ticker (...-TRS---...)	src/nport/custodian.py:733
notionalAmt	Custodian	MarketValue = shares × price — the contract size	src/nport/custodian.py:734
unrealizedAppr	EagleSTAR	PVal Total Unreal G/L Base, matched on Primary Asset ID	src/nport/eaglestar.py:231
counterpartyName / counterpartyLei	Human review	swap_counterparties sheet — ticker code → legal name + GLEIF LEI	src/nport/master_sheet.py:773
recFixedOrFloating / recDesc	Human review	swap_legs sheet — from the trade confirm	src/nport/master_sheet.py:758
pmntFloatingRtIndex / pmntFloatingRtSpread / pmntRateTenor / pmntRateUnit	Human review	swap_legs sheet — from the trade confirm	src/nport/master_sheet.py:758
refCusip	Custodian	Leading segment of the swap ticker	src/nport/custodian.py:740
refIsin / refTicker / refIssuerName / refIssueTitle	Bloomberg	=BDP(refCusip Equity, ...) on the reference security	src/nport/master_sheet.py:190

FUND FINANCIALS (B.1)			
totAssets	EagleSTAR	Trial Balance total assets	src/nport/eaglestar.py:307
totLiabs	EagleSTAR	Trial Balance total liabilities	src/nport/eaglestar.py:309
netAssets	EagleSTAR	Trial Balance net assets (overrides the custodian figure)	src/nport/eaglestar.py:311
amtPayOneYrOther	EagleSTAR	Trial Balance payables	src/nport/eaglestar.py:304
assetsAttrMiscSec / assetsInvested / delayDeliv / standByCommit / liquidPref	Tool constant	0 — a genuine zero for a plain ETF, not a placeholder	src/nport/filing_master.py:72

amtPayOneYr* / amtPayAftOneYr* (other lines)	Tool constant	0 — no borrowings or affiliate payables	src/nport/filing_master.py:72
RETURNS (B.5)			
rtn1 / rtn2 / rtn3	Bloomberg	=BDP total return over each reporting month; N/A before inception	src/nport/filing_master.py:45
GAINS (B.5)			
netRealizedGainMon1-3	EagleSTAR	Trial Balance month-end deltas	src/nport/eaglestar.py:300
netUnrealizedApprMon1-3	EagleSTAR	Trial Balance month-end deltas	src/nport/eaglestar.py:301
FLOWS (B.2)			
mon1-3Sales / mon1- 3Redemption	EagleSTAR	Trial Balance subscriptions/redemptions. AP order book cross-checks only	src/nport/eaglestar.py:337
mon1-3Reinvestment	Tool constant NO FEED	Not in an order book — needs the transfer agent	src/nport/filing_master.py:80
RISK (B.3)			
durAdj / spreadDur	Bloomberg	=BDP DUR_ADJ_MID / OAS_SPREAD_DUR_MID per debt holding	src/nport/filing_master.py:50
maturity / ratingSP	Bloomberg	=BDP MATURITY / RTG_SP, bucketed by tenor	src/nport/filing_master.py:50
INDEX (B.6)			
nameDesignatedIndex / indexIdentifier	Human review	designated_index sheet — from the prospectus	src/nport/humanreview.py:91
ADMIN			
submissionType / isFinalFiling / isNonCashCollateral	Tool constant	NPORT-P / N / N	src/nport/filing_master.py:62
liveTestFlag	Tool constant	TEST until you flip it; LIVE is gated on a clean reconciliation	src/nport/cli.py:878
repPdEnd / repPdDate	Derived	Last calendar day of the filing period	src/nport/custodian.py:805
dateSigned	Derived	Set from the period; override in the workbook	src/nport/filing_master.py:115

Verified at generation time

- 65 rows: code anchor located for each
- Bloomberg claims checked against BLOOMBERG_SPECS (10 cols) + risk sheet (4 cols)
- EagleSTAR claims checked against 17 writer keys
- Human-review claims checked against 15 review value columns
- Custodian claims checked against 67 transform-assigned keys
- filing_master HEADER: 45 columns, 0 unmapped
- Coverage measured over **1,996 holdings** across **104 funds** in period 2026-06